APPLIED SCIENCE PRIVATE UNIVERSITY (ASU)

FACULTY OF INFORMATION TECHNOLOGY (FIT)

Final Lab Exam 2024-3

جامعة العلوم التطبيقية الخاصة
كلية تكنولوجيا المعلومات
تقييم امتحان عملي نهائي ٢٠٢٤-٣

Subject: *Object-Oriented Programming (1) 1301108*

Instru[redacted]

Date: Sep. 1, 2025

Time: 2:30 AM

Period :60 minutes

Studen[redacted]

Student [redacted]

☐ CS ☐ SE ☐ CCC [redacted]

PLEASE NOTE THAT THIS LAB QUIZ IS 2 PAGES AND WITH A TOTAL OF 10 POINTS.

Exam Instructions:

- Use NetBeans IDE to solve the questions below.

Question 1

8.5

[10 POINTS]

A. [2 marks] Define an interface named Measurable that declares:

- A method getMeasurement() that returns a double representing some measurable value.

B. [3.5 marks] Write an abstract class named Shape. The class should have:

- Two private attributes: shapeName (type String) and color (type String).
- A constructor to initialize shapeName and color.
- An abstract method named calculateArea() that returns a double representing the area of the shape.
- Accessor (getter) methods for shapeName and color.
- A final method named getShapeSummary() that returns a String in the format:
  "Shape: [shapeName], Color: [color]"

Note: Do not add mutators or a toString() method to the Shape class.

C. [3.5 marks] Write a class named Rectangle that:

- Inherits from the Shape class.
- Implements the Measurable interface.
- Has two private attributes:
  - length (type double).
  - width (type double).
- A constructor to initialize shapeName, color, length, and width.
- An implementation of calculateArea() that returns length × width.
- An implementation of getMeasurement() that returns the perimeter of the rectangle using the formula:
  Perimeter = 2 × (length + width)
- A toString() method that returns a string containing the rectangle details in a readable format.

D. [1 mark] Write a ShapeDemo class with a main method to:

- Create a Rectangle object with the following values:
  - shapeName: "Rectangle"
  - color: "Blue"
  - length: 5.0
  - width: 3.0
- Use calculateArea(), toString(), and getMeasurement() to display the area, shape details, and perimeter.

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```
① package quiz1;
public class Server {
        String serverName;
        int cpuLoad;  2
        String serverType;

    public Server (String serverName, int cpuLoad, String serverType){
            this.serverName = serverName;
            this.cpuLoad = cpuLoad;
            this.serverType = serverType;
}public double calculateRemainingCapacity() {
    if(serverType.equalsIgnorCase("Web") {
        return remaining capacity = 100-(cpuLoad*1.1);
}else if (serveType.equalsIgnoreCase("Database") {
        return remaining capacity = 100-(cpuLoad*1.3);
}else
  {return 0;
}public void printDetails() {
    System.out.print("Server Name:"+ this.serverName);
    System.out.print("cpuLoad:"+ this.cpuLoad);
    System.out.print("Server Type:"+ this.serverType);
  }
}
}
```

1.75

3

Good luck

② package quiz1;

import java.util.Scanner;

```
public class ServerTest{
public static void main (String [] args) {
 Scanner in = new Scanner(System.in);
 System.out.print(" Server Name : ");
   String a = in.next();
System.out.print(" cpu load :");
  int b = in.nextInt();
System.out.print(" Server Type :");
   String c = in.next();
Server obj = in . Server(a,b,c);
System.out.print( obj.calculateRemainingCapacit());
}
```

3.25